



Martin Thuo holds a vial of the liquid-metal particles produced by his research group (Image courtesy: www.news.lastate.edu)

Tiny droplets of liquid metal were covered with a thin, uniform coating, to form stable particles of the under-cooled liquid metal. The researchers wanted to build a system where the metals do not turn into solids.

Solid batteries can improve safety

Lithium-ion batteries store a lot of energy in a small space, making these the energy source of choice for mobile electronic devices. Researchers at ETH Zurich, Switzerland, have developed a type of battery that, unlike conventional ones, consists entirely of solid chemical compounds (contains neither liquids nor gels) and is non-flammable. The battery cannot ignite, even at very high temperatures, giving it a safety advantage over conventional batteries. In addition, it allows new forms of battery design.

In conventional lithium-ion batteries, as well in most other batteries, the positive and negative poles (the two electrodes) are made of solid conductive compounds; charges move between these electrodes in a liquid or gel electrolyte. If you charge such a battery improperly, or leave it sitting out in the sun, the liquid can ignite or the gel can swell up.

This is not the case with solid-state batteries. In these types of batteries, both the electrodes and the intermediary electrolyte are made of solid material. Solid electrolytes do not catch fire even when heated to high temperatures or exposed to the air.



A slice of (white) lithium garnet electrolyte coated with (black) lithium compound acting as the battery's minus pole (Image courtesy: ETH Zurich)

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM and LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed intensity to allow for natural LED intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT

Binay Opto Electronics Pvt. Ltd.

27, PANDITIA TERRACE, Calcutta 700 029, India
Telephone: +91 (33) 4006 9875, 2475 0392, 2475 0030
email: info@binayled.com, <http://www.binayled.com>

POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983